

1) Apply gram Schmidt process to

the vectors $u_1 = (2 -1 2)$ $u_2 = (1 0 1)$

and $u_3 = (0 2 1)$, to transform them to orthonormal vectors.

Proof:-

$$v_1 = u_1 = (2, -1, 2)$$

$$v_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{4}{9} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/9 \\ 4/9 \\ 1/9 \end{bmatrix}$$

$$v_3 = u_3 - \frac{\langle u_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle u_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$$

$$= \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} - 0 - \frac{9}{2} \begin{bmatrix} 1/9 \\ 4/9 \\ 1/9 \end{bmatrix} = \begin{bmatrix} 0 - 1/2 \\ 2 - 2 \\ 1 - 1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix}$$

→ IM

$$w_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{4+4+1}} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -1/3 \\ 2/3 \end{bmatrix} \rightarrow \text{IM}$$

$$w_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{\frac{1}{81} + \frac{16}{81} + \frac{1}{81}}} \begin{bmatrix} 1/9 \\ 4/9 \\ 1/9 \end{bmatrix} \rightarrow \text{IM}$$

$$= \frac{3}{\sqrt{2}} \begin{bmatrix} 1/9 \\ 4/9 \\ 1/9 \end{bmatrix} = \begin{bmatrix} 1/3\sqrt{2} \\ 4/3\sqrt{2} \\ 1/3\sqrt{2} \end{bmatrix}$$

$$w_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{\frac{1}{4} + 0 + \frac{1}{4}}} \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \frac{2}{\sqrt{2}} \begin{bmatrix} -1/2 \\ 0 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \rightarrow \text{IM}$$

Orthonormal vectors are

$$\left\{ \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3} \right), \left(\frac{1}{3\sqrt{2}}, \frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) \right\}$$

2) Find the QR factorization of the

$$\text{matrix } A = \begin{bmatrix} -4 & 6 \\ 3 & -5 \end{bmatrix}$$

$$\text{Here } v_1 = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$

$$\frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{16+9}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix} \rightarrow \text{IM}$$

$$v_2 = \begin{bmatrix} 6 \\ -5 \end{bmatrix} - \frac{-39}{25} \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ -5 \end{bmatrix} - \begin{bmatrix} \frac{156}{25} \\ -\frac{117}{25} \end{bmatrix}$$

$$= \begin{bmatrix} -6/25 \\ -8/25 \end{bmatrix}$$

$$\frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{\frac{36+64}{625}}} \begin{bmatrix} -6/25 \\ -8/25 \end{bmatrix} = \frac{5}{2} \begin{bmatrix} -6/25 \\ -8/25 \end{bmatrix} = \begin{bmatrix} -3/5 \\ -4/5 \end{bmatrix}$$

$\rightarrow \text{IM}$

$$\therefore Q = \begin{bmatrix} -4/5 & -3/5 \\ 3/5 & -4/5 \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} -4/5 & 3/5 \\ -3/5 & -4/5 \end{bmatrix} \begin{bmatrix} -4 & 6 \\ 3 & -5 \end{bmatrix} \rightarrow 1M$$

$$R = \begin{bmatrix} 5 & -39/5 \\ 0 & 2/5 \end{bmatrix} \rightarrow 1M$$

3) Find the singular values of the matrix $B = \begin{bmatrix} 1 & 2 & 2 \\ -2 & 2 & -1 \end{bmatrix}$

$$B^T B = \begin{bmatrix} 1 & -2 \\ 2 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ -2 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

$\rightarrow 1/2 M$

$$|B^T B - \lambda I| = \begin{vmatrix} 5-\lambda & -2 & 4 \\ -2 & 8-\lambda & 2 \\ 4 & 2 & 5-\lambda \end{vmatrix} = 0 \quad \rightarrow \frac{1}{2} M$$

$$(5-\lambda)((8-\lambda)(5-\lambda) - 4) + 2(-10 + 2\lambda - 8) + 4(-4 - 32 + 4\lambda) = 0$$

$$\Rightarrow (5-\lambda)(40 + \lambda^2 - 13\lambda - 4) + 2(2\lambda - 18) + 4(4\lambda - 36) = 0$$

$$\Rightarrow (5-\lambda)(\lambda^2 - 13\lambda + 36) + 4(\lambda - 9) + 16(\lambda - 9) = 0$$

$$\Rightarrow (5-\lambda)(\lambda - 9)(\lambda - 4) + 4(\lambda - 9) + 16(\lambda - 9) = 0$$

$$\Rightarrow (\lambda - 9)(-\lambda^2 + 9\lambda) = 0$$

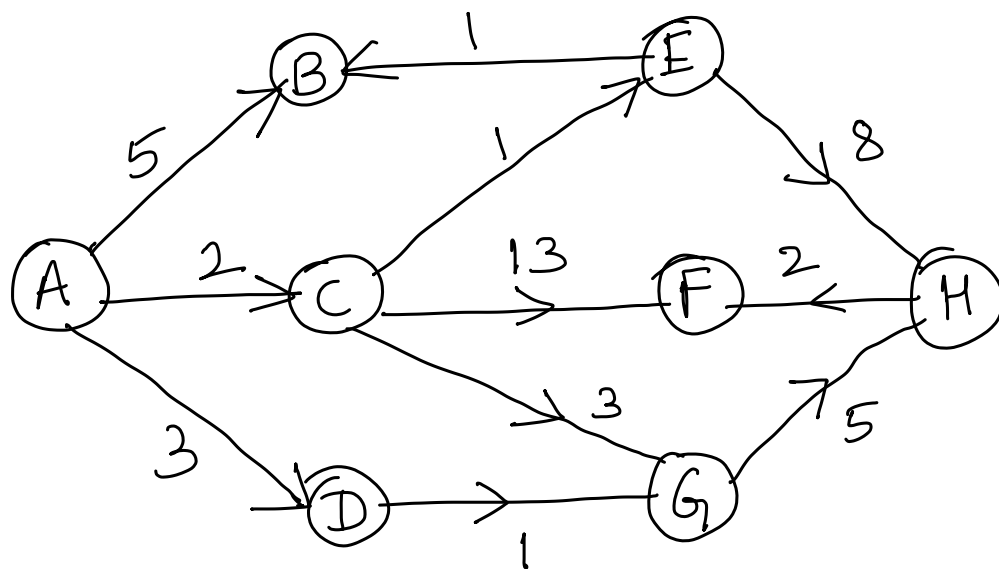
$$\Rightarrow (\lambda - 9)(\lambda)(-\lambda + 9) = 0$$

$$\Rightarrow \lambda = 9, 9, 0$$

$\therefore$ Eigenvalues are $\lambda_1 = 9, \lambda_2 = 9, \lambda_3 = 0$

$\Rightarrow$ Singular values are $3, 3, 0 \rightarrow 1M$

4) Apply dijkstra's algorithm to the following graph G to find the shortest path from vertex A to all the vertices of G .



$$A \xrightarrow{2} C \longrightarrow \frac{1}{2} M$$

$$A \xrightarrow{2} C \xrightarrow{1} E \longrightarrow \frac{1}{2} M$$

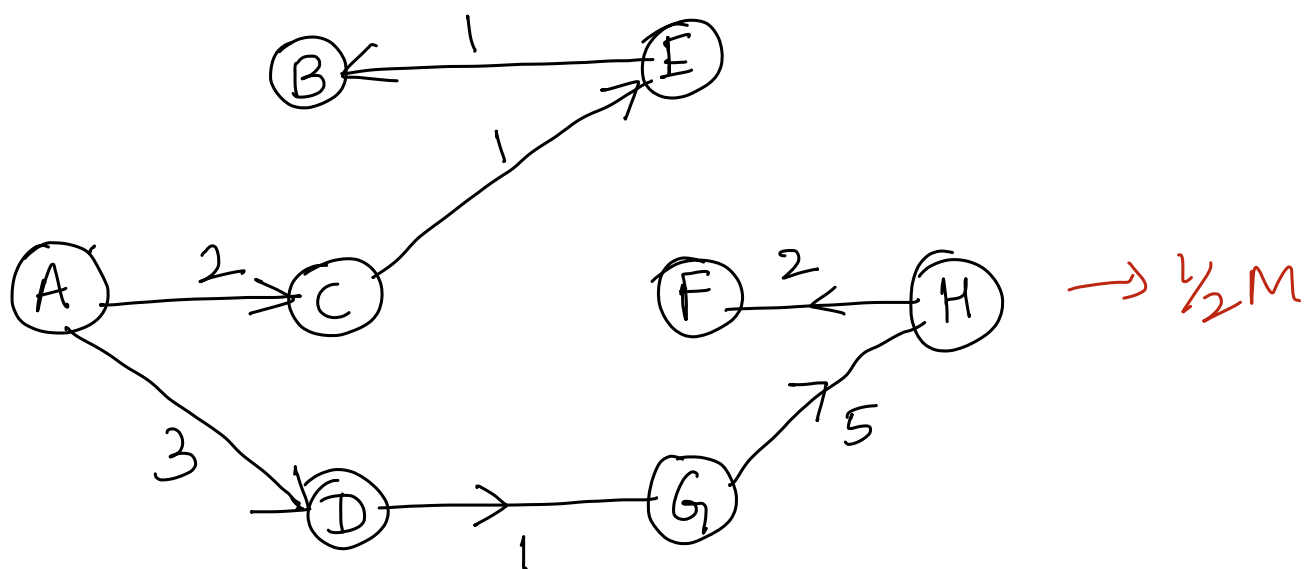
$$A \xrightarrow{2} C \xrightarrow{1} E \xrightarrow{1} B \longrightarrow \frac{1}{2} M$$

$$A \xrightarrow{3} D \longrightarrow \frac{1}{2} M$$

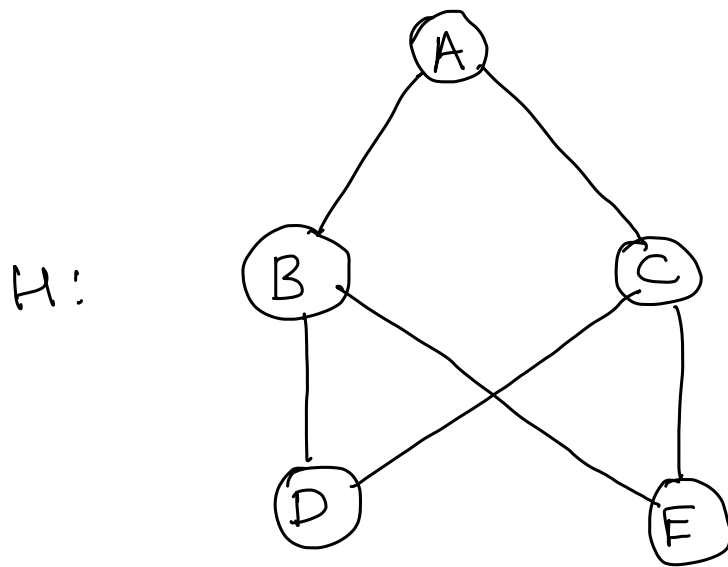
$$A \xrightarrow{3} D \xrightarrow{1} G \longrightarrow \frac{1}{2} M$$

$A \xrightarrow{3} D \xrightarrow{1} G \xrightarrow{5} H \rightarrow \frac{1}{2} M$

$A \xrightarrow{3} D \xrightarrow{1} G \xrightarrow{5} H \xrightarrow{2} F \rightarrow \frac{1}{2} M$



5) Find the number of spanning trees of the following graph H. using Kirchhoff's matrix - tree theorem.



Degree matrix $D =$

$$\begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \frac{1}{2}M$$

Adjacency matrix $A =$

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \rightarrow \frac{1}{2}M$$

Laplacian matrix $L = D - A$

$$L = \begin{bmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & 0 & -1 & -1 \\ -1 & 0 & 3 & -1 & -1 \\ 0 & -1 & -1 & 2 & 0 \\ 0 & -1 & -1 & 0 & 2 \end{bmatrix} \rightarrow \frac{1}{2} M$$

$$\text{Cofactor } C_{22} = (-1)^{2+2} \begin{vmatrix} 2 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{vmatrix} \rightarrow \frac{1}{2} M$$

$$= 2 \begin{vmatrix} 3 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{vmatrix} + \begin{vmatrix} -1 & -1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= 2 \{ 3(4) + 1(-2) - 1(2) \} - 1(4)$$

$$= 2(12 - 2 - 2) - 4$$

$$= 12 \rightarrow 1 M$$

6) Solve $(x^2+1)y'' + (x+2)y' - 2xy = 4$,

$y(0)=0, y(1)=2$ with $h=0.5$ by

finite difference method.

Solⁿ: $(x_i^2+1)y_i'' + (x_i+2)y_i' - 2x_i y_i = 4$

$$\begin{array}{ccc} y_0=0 & y_1=? & y_2=2 \\ | & | & | \\ x_0=0 & x_1=0.5 & x_2=1 \end{array}$$

→ 1/2 M

$$(x_i^2+1) \left[\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} \right] + (x_i+2) \left[\frac{y_{i+1} - y_{i-1}}{2h} \right] - 2x_i y_i = 4$$

Put $h=0.5 = 1/2 \Rightarrow h^2 = 1/4$

$$4(x_i^2+1) [y_{i+1} - 2y_i + y_{i-1}] + (x_i+2) [y_{i+1} - y_{i-1}] - 2x_i y_i = 4,$$

$i=1 \rightarrow 1M$

Put $i=1$

$$4(x_1^2+1)(y_2 - 2y_1 + y_0) + (x_1+2)(y_2 - y_0) - 2x_1 y_1 = 4$$

→ 1/2 M

$$\Rightarrow 4(0.5^2+1)(2 - 2y_1 + 0) + (0.5+2)(2 - 0) - 2 \times 0.5 \times y_1 = 4$$

$$\Rightarrow 5(2 - 2y_1) + 5 - y_1 = 4$$

$$\Rightarrow 10 - 10y_1 + 5 - y_1 - 4 = 0$$

$$\Rightarrow -11y_1 + 11 = 0$$

$$\Rightarrow y_1 = 1$$

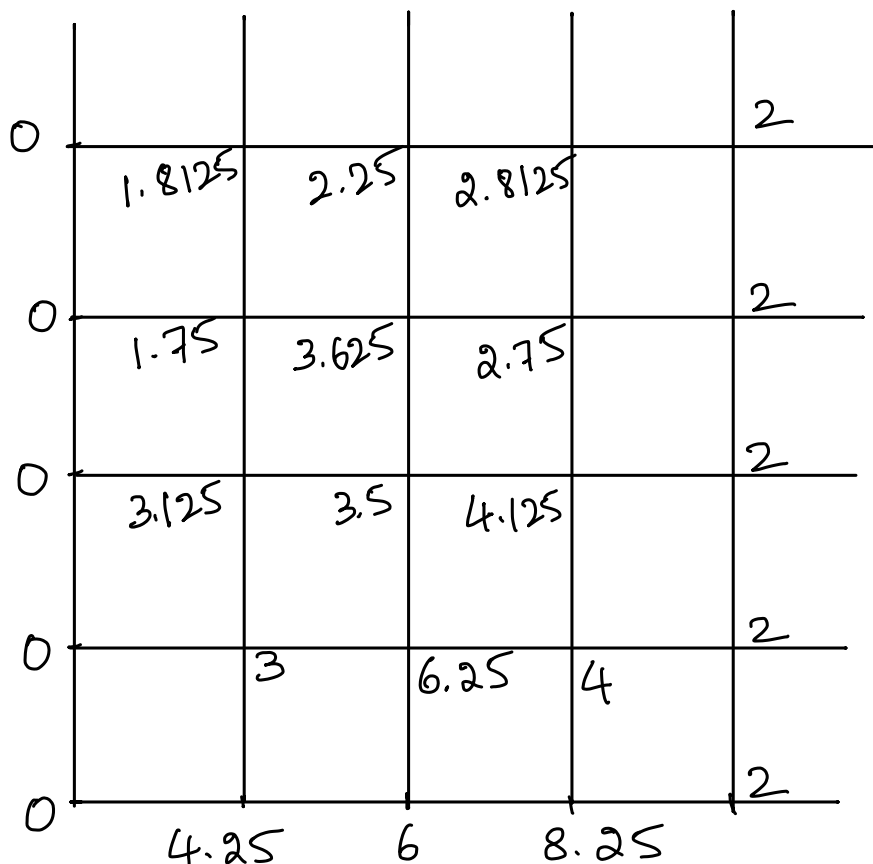
$$\therefore y(0.5) = 1 \quad \rightarrow 1M$$

7) Compute u for four time steps with
 $h = 0.5$, $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$, $0 < x < 2$, $t > 0$

$$u(x, 0) = x^2 + 2x + 3$$

$$u(0, t) = 0$$

$$u(2, t) = 2$$



Given $u(x,0) = x^2 + 2x + 3$

(Put $x = \frac{1}{2}$) $u_{10} = (\frac{1}{2})^2 + 2(\frac{1}{2}) + 3 = 4.25$

(Put $x = 1$) $u_{20} = 1^2 + 2(1) + 3 = 6$

(Put $x = \frac{3}{2}$) $u_{30} = (\frac{3}{2})^2 + 2(\frac{3}{2}) + 3 = 8.25$
 $\rightarrow 1M$

$$u_{i,j+1} = \frac{u_{i+1,j} + u_{i-1,j}}{2}$$

$u_{00} = 0, u_{10} = 4.25, u_{20} = 6, u_{30} = 8.25, u_{40} = 2$

$u_{01} = 0, u_{11} = 3, u_{21} = 6.25, u_{31} = 4, u_{41} = 2$ $\left. \vphantom{u_{01}} \right\} 1\frac{1}{2}M$

$u_{02} = 0, u_{12} = 3.125, u_{22} = 3.5, u_{32} = 4.125, u_{42} = 2$

$u_{03} = 0, u_{13} = 1.75, u_{23} = 3.625, u_{33} = 2.75, u_{43} = 2$ $\left. \vphantom{u_{03}} \right\} 1\frac{1}{2}M$

$u_{04} = 0, u_{14} = 1.8125, u_{24} = 2.25, u_{34} = 2.8125, u_{44} = 2$

8) Following is the payoff matrix for a two person zero sum game. Using minimax criterion, find the best strategy for each player.

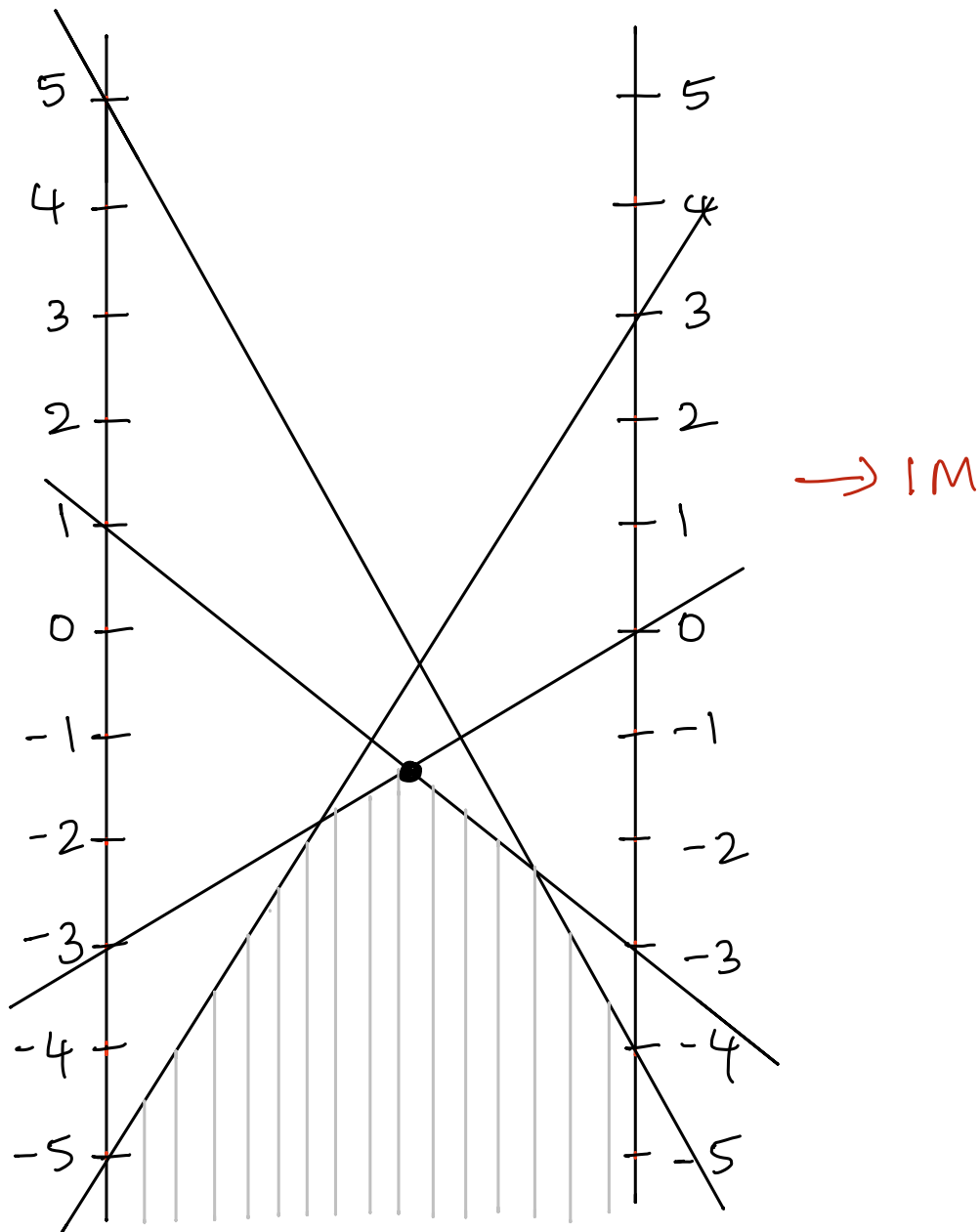
		Player B				
		B1	B2	B3	B4	
Player A	A1	3	-5	-2	7	-5
	A2	1	-2	3	5	<u>-2</u>
	A3	2	-6	4	3	-6
		3	<u>-2</u>	4	7	
			minimax			
						maximin

Here $\max_{\min} \underline{v} = \min_{\max} \bar{v} = -2 \rightarrow \text{IM}$

$\therefore$ There exists a saddle point. The value of the game is -2 , which says A always loses 2 and optimal strategy is $(A2, B2)$. $\rightarrow \text{IM}$

9) Use graphical method to reduce the following game and hence solve

		Player B			
		y_1	y_2	y_3	y_4
Player A	x_1	-3	-4	0	3
	$1-x_1$	1	5	-3	-5



The payoff matrix is reduced to

$$\begin{array}{cc} & y_1 & y_3 = 1 - y_1 \\ \begin{array}{c} x_1 \\ 1 - x_1 \end{array} & \begin{bmatrix} -3 & 0 \\ 1 & -3 \end{bmatrix} & y_2 = y_4 = 0 \end{array} \rightarrow \frac{1}{2} M$$

$$-3x_1 + 1 - x_1 = -3(1 - x_1)$$

$$\Rightarrow -4x_1 + 1 = 3x_1 - 3$$

$$\Rightarrow 7x_1 = 4$$

$$\Rightarrow x_1 = 4/7 \quad \text{and} \quad x_2 = 1 - x_1 = 1 - 4/7 = 3/7$$

$$-3y_1 = y_1 - 3(1 - y_1)$$

$$\Rightarrow -3y_1 = y_1 - 3 + 3y_1$$

$$\Rightarrow 7y_1 = 3$$

$$\Rightarrow y_1 = 3/7 \Rightarrow y_3 = 1 - y_1 = 1 - 3/7 = 4/7$$

$\therefore$ Optimum strategy for A is $(\frac{4}{7}, \frac{3}{7}) \rightarrow \frac{1}{2} M$

and optimum strategy for B is $(\frac{3}{7}, 0, \frac{4}{7}, 0) \rightarrow \frac{1}{2} M$

$$\text{The value of the game} = -4\left(\frac{4}{7}\right) + 1 = \frac{-9}{7},$$

which means A always loses $\frac{9}{7}$.

$\rightarrow \frac{1}{2} M$

